Zheng YiJie

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PROFESSIONAL SUMMARY

Computer Science undergraduate at NUS focused on building end-to-end software systems that solve real business problems. Experienced in designing and deploying automation and data-driven solutions with measurable impact (up to 80% workload reduction). Strong interest in AI agents, backend systems, and working directly with users to scope, build, and iterate quickly in ambiguous environments.

EDUCATION

Bachelor Of Computer Science **2024 - 2028**
National University of Singapore Singapore
Relevant Coursework | Software Engineering, Artificial Intelligence, Database Systems, Data Structures & Algorithm

WORK EXPERIENCE

Operations Engineer **Dec 2023 – Jul 2024**
Liberty Specialty Markets Singapore

- Designed and deployed automation solutions to handle GST-related invoice processing, improving data accuracy during system-wide tax updates.
- Identified inefficiencies in financial workflows and built Python-based tools to reduce manual intervention and improve processing reliability.
- Automated daily reconciliation pipelines using Excel VBA, reducing turnaround time and improving consistency of reporting.

Data Analyst (CFC) **Oct 2022 – Nov 2023**
Ministry Of Defence Singapore

- Led the design and implementation of an automated account registration system within SAF infrastructure, replacing a manual workflow and reducing administrative workload by 80%.
- Worked with stakeholders to identify operational bottlenecks and translate requirements into scalable system solutions.
- Debugged and optimized workflows to ensure reliability and smooth system rollout.

PROJECTS

AI Workflow Automation Tool (Invoice Processing Agent)

- Built an AI-powered backend system to extract and process structured data from invoices and emails
- Designed a pipeline integrating LLM-based extraction with PostgreSQL storage and API endpoints
- Implemented automated workflows for classification, summarisation, and data persistence
- Structured outputs to handle edge cases and improve reliability of AI-generated results

DSTA BrainHack Finalist **Dec 2025**

- Collaborated in cross-functional teams to design and prototype technical solutions under time constraints, contributing to system logic and implementation.
- Presented solutions to judges, clearly communicating technical decisions and real-world applicability.

NUS Orbital CP2106 – Unity Game Development | C#, Unity **May 2025**

- Designed and implemented modular gameplay systems using object-oriented architecture, improving scalability and maintainability.
- Worked in a 2-person team to deliver features iteratively, managing task ownership and code reviews.
- Built core systems including player movement, obstacle generation, and scoring logic.

TECHNICAL SKILLS

Languages | Python, JavaScript, Java, C/C++, SQL, C#, Excel VBA, HTML/CSS
Frameworks | React, Node.js, Next.js
Developer Tools | Git, GitHub, Visual Studio Code, PyCharm, IntelliJ IDEA, Unity Editor, Docker, PostgreSQL
Libraries | NumPy